

About This Document

Statistics and probability are core to the data-scientist assessment and interview (p-values, CLT, error types, distributions).

Master these fundamentals; they recur in both the MCQ round and the technical interview.

Descriptive Statistics

9 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Find the mean of 2, 4, 6, 8, 10.

(A) 5

(B) 6 ✓

(C) 7

(D) 8

Answer: (B) 6

Q2. Find the median of 3, 1, 4, 1, 5, 9, 2.

(A) 2

(B) 3 ✓

(C) 4

(D) 1

Answer: (B) 3

Q3. Find the mode of 2, 3, 3, 5, 7, 3, 8.

(A) 2

(B) 3 ✓

(C) 5

(D) 7

Answer: (B) 3

Q4. Standard deviation is a measure of:

(A) Central tendency

(B) Dispersion ✓

(C) Correlation

(D) Skewness

Answer: (B) Dispersion

Q5. Variance is equal to:

(A) The standard deviation

(B) The square root of the SD

(C) The square of the standard deviation ✓

(D) The mean of deviations

Answer: (C) The square of the standard deviation

Q6. Which measure of central tendency is most affected by extreme outliers?

- (A) Mode
- (B) Median
- (C) Mean ✓**
- (D) Range

Answer: (C) Mean

Q7. In a right-skewed distribution, the usual order is:

- (A) Mean < Median < Mode
- (B) Mean > Median > Mode ✓**
- (C) Mean = Median = Mode
- (D) Median > Mean > Mode

Answer: (B) Mean > Median > Mode

Q8. The median corresponds to which percentile?

- (A) 25th
- (B) 50th ✓**
- (C) 75th
- (D) 100th

Answer: (B) 50th

Q9. The interquartile range (IQR) is defined as:

- (A) $Q3 - Q1$ ✓**
- (B) $Q3 + Q1$
- (C) Max - Min
- (D) $Q2 - Q1$

Answer: (A) $Q3 - Q1$

Probability

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, then $P(A \text{ and } B)$ is:

- (A) 0.8
- (B) 0.15 ✓**
- (C) 0.2
- (D) 0.5

Answer: (B) 0.15

Q2. For mutually exclusive events with $P(A) = 0.2$ and $P(B) = 0.3$, $P(A \text{ or } B)$ is:

- (A) 0.06
- (B) 0.5 ✓**
- (C) 0.1
- (D) 0.25

Answer: (B) 0.5

Q3. The expected value of a single roll of a fair six-sided die is:

(A) 3

(B) 3.5 ✓

(C) 4

(D) 2.5

Answer: (B) 3.5

Q4. Two fair coins are tossed. Probability of getting exactly one head:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$ ✓

(C) $\frac{3}{4}$

(D) $\frac{1}{3}$

Answer: (B) $\frac{1}{2}$

Q5. If events A and B are independent, then $P(A | B)$ equals:

(A) $P(B)$

(B) $P(A)$ ✓

(C) $P(A \text{ and } B)$

(D) 0

Answer: (B) $P(A)$

Q6. The sum of the probabilities of all outcomes in a sample space is:

(A) 0

(B) 0.5

(C) 1 ✓

(D) Depends on the events

Answer: (C) 1

Q7. Bayes' theorem is used to:

(A) Compute variance

(B) Relate $P(A|B)$ to $P(B|A)$ ✓

(C) Test hypotheses

(D) Measure correlation

Answer: (B) Relate $P(A|B)$ to $P(B|A)$

Q8. A card is drawn from a standard deck. Probability that it is a heart:

(A) $\frac{1}{13}$

(B) $\frac{1}{4}$ ✓

(C) $\frac{1}{52}$

(D) $\frac{1}{2}$

Answer: (B) $\frac{1}{4}$

Inferential Statistics and Hypothesis Testing

10 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. A p-value less than the significance level (e.g. 0.05) suggests you should:

(A) Accept the null hypothesis

(B) Reject the null hypothesis ✓

(C) Ignore the result

(D) Increase the sample size

Answer: (B) Reject the null hypothesis

Q2. A Type I error is:

(A) Failing to reject a false null hypothesis

(B) Rejecting a true null hypothesis ✓

(C) Accepting the alternative correctly

(D) A sampling error

Answer: (B) Rejecting a true null hypothesis

Q3. A Type II error is:

(A) Rejecting a true null hypothesis

(B) Failing to reject a false null hypothesis ✓

(C) A measurement error

(D) A rounding error

Answer: (B) Failing to reject a false null hypothesis

Q4. The Central Limit Theorem states that the sampling distribution of the mean:

(A) Is always uniform

(B) Approaches a normal distribution as sample size increases ✓

(C) Has zero variance

(D) Equals the population distribution

Answer: (B) Approaches a normal distribution as sample size increases

Q5. For a normal distribution, approximately what percent of data lies within one standard deviation of the mean?

(A) 50%

(B) 68% ✓

(C) 95%

(D) 99.7%

Answer: (B) 68%

Q6. For a normal distribution, approximately what percent of data lies within two standard deviations of the mean?

(A) 68%

(B) 90%

(C) 95% ✓

(D) 99.7%

Answer: (C) 95%

Q7. Pearson's correlation coefficient ranges between:

(A) 0 and 1

(B) -1 and 1 ✓

(C) -infinity and infinity

(D) 0 and 100

Answer: (B) -1 and 1

Q8. A correlation coefficient of 0 indicates:

(A) Perfect positive relation

(B) Perfect negative relation

(C) No linear relationship ✓

(D) A causal relationship

Answer: (C) No linear relationship

Q9. As sample size increases, the standard error of the mean generally:

(A) Increases

(B) Decreases ✓

(C) Stays constant

(D) Becomes negative

Answer: (B) Decreases

Q10. Which test is commonly used to compare the means of two groups?

(A) Chi-square test

(B) t-test ✓

(C) ANOVA of one group

(D) Correlation test

Answer: (B) t-test

Tredence

Statistics & Probability

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

Version 1.0 • Last Updated: July 2026

Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

Table of Contents

About This Document	1
Descriptive Statistics	1
Probability	2
Inferential Statistics and Hypothesis Testing	3